Final Exam

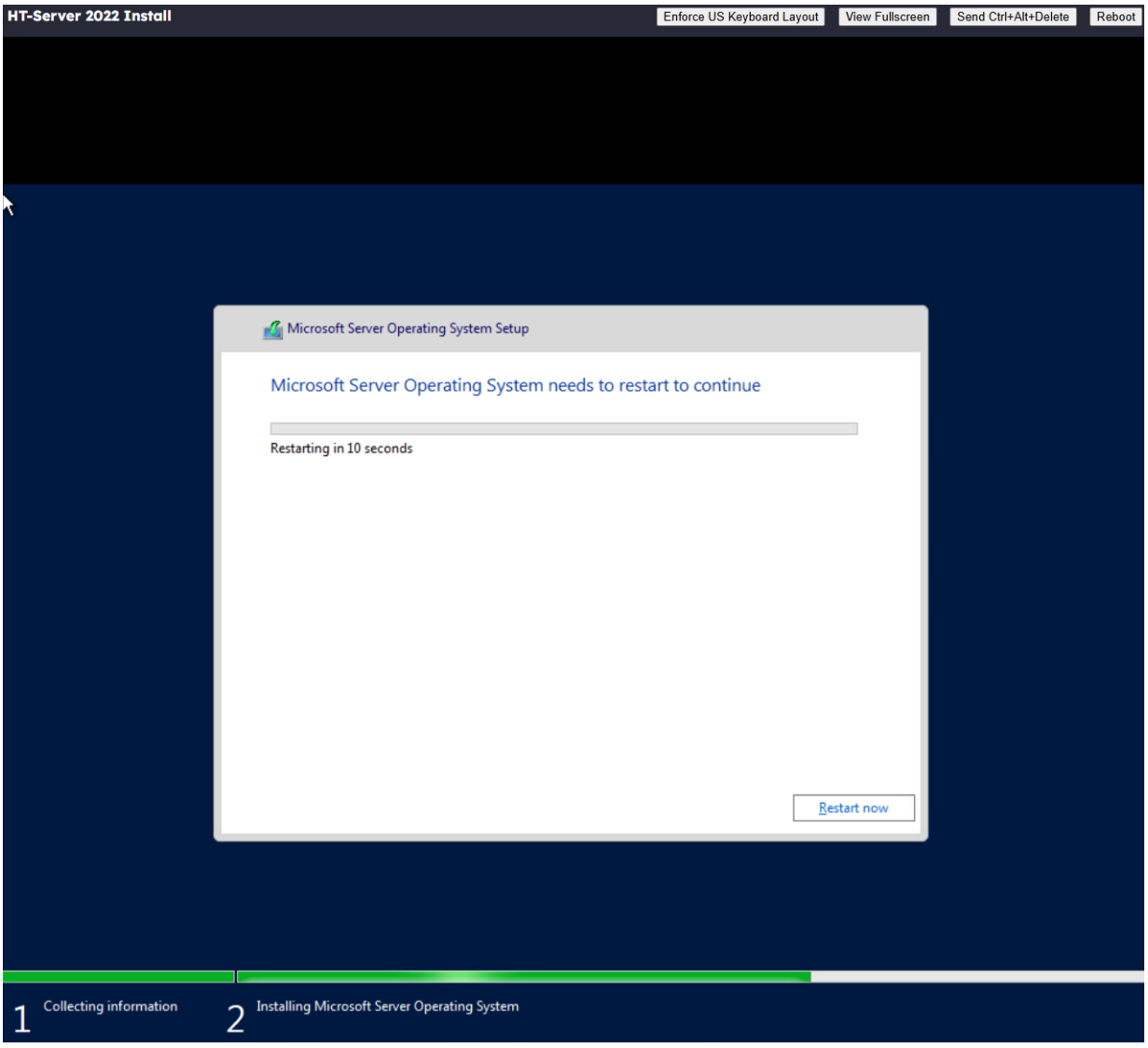
Name **Blong Lor** _____ **/100**

Perform all the following tasks logged in to your computer with the **Administrator** account using the password as **Htcfun2020** in your own root domain that you will create. Be sure to read all directions before proceeding as there are shortcuts to accomplishing the steps below if careful planning is used. Include at least **one screenshot (output)** for each question and then upload it in to D2L's final test Dropbox.

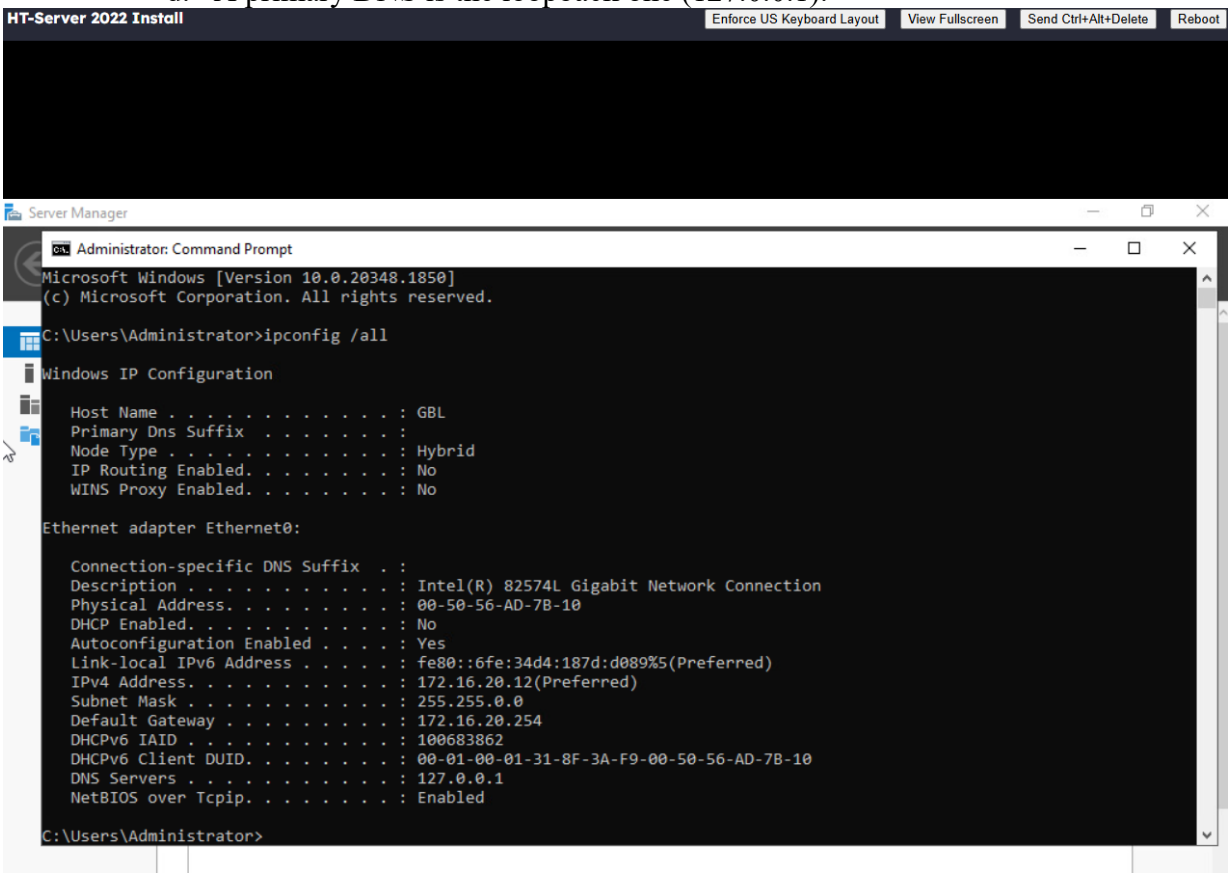
Instructions: Step 1: Click on the following link and sign in on the **ACI/HTC Cyber Range** portal.
https://lab.infoseclearning.com/user/login?destination=/course/G67EQ3DDUA/lab/HQV33RBR8V

Step 2: Select the **Hennepin Tech: Cyber Range A**
Step 3: Click the **"Erase and Start Over"** option to reset the Server 2022 virtual machine.
Step 4: Hit the red **Start** button in the center to start your **HTC Cyber Range** Virtual Machines.
Step 5: Find and double-click the **Server 2022 Install** icon and then do the following tasks:

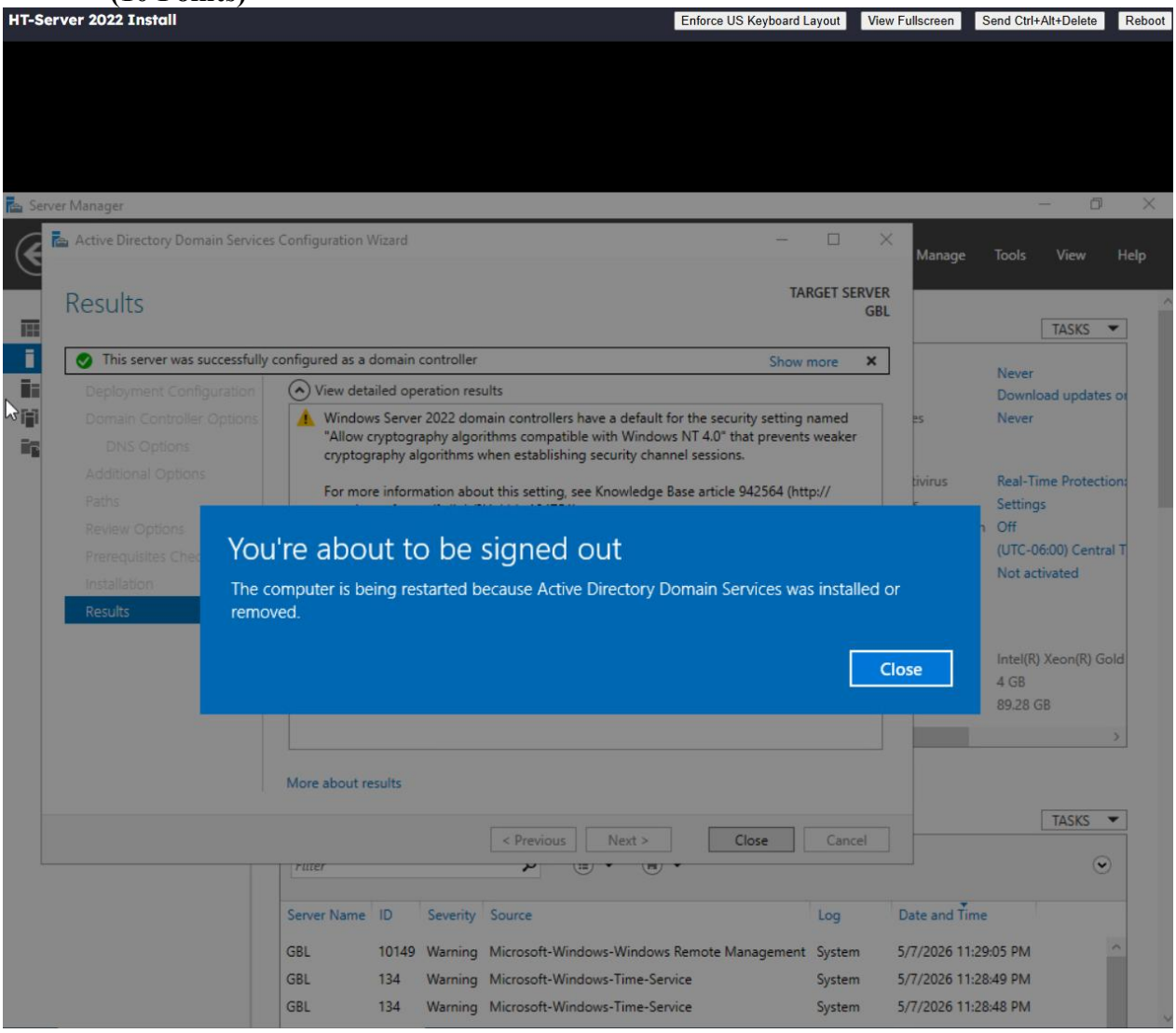
1. Install Windows 2022 Datacenter server (GUI) on your VM (one big partition).
   No Product Key is required for **Datacenter 2022**. **(10 Points)**



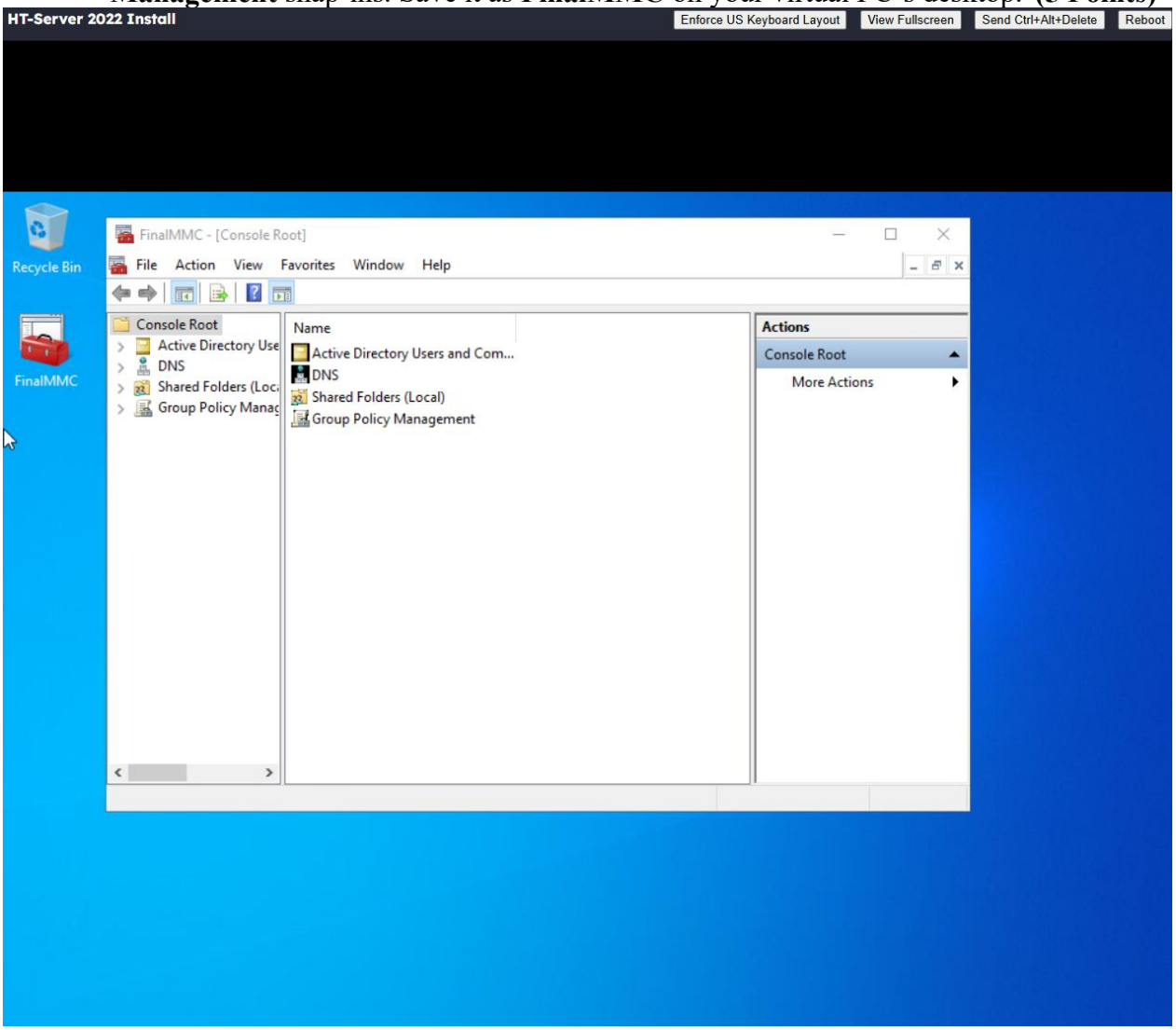
2. After logging in the first time, make your server's Network Card to have a static IP address with the following parameters: **(5 Points)**
    a. IP address of 172.16.20. **xx** (where **xx** is a number between 1 – 24).
    b. A subnet mask of 255.255.0.0.
    c. A default gateway of 172.16.20.254
    d. A primary DNS is the loopback one (127.0.0.1).



3. Install Active Directory (AD) with DNS with a domain name of "**test*xx*.lab**" (where **xx** is your last name) just like we have done in class. You may choose the location of any AD database locations yourself or take the defaults. You need not worry about backwards compatibility with older servers. **(10 Points)**



4. Create an MMC with the **AD Users and Computers**, **DNS**, **Shared Folders**, and **Group Policy Management** snap-ins. Save it as **FinalMMC** on your virtual PC's desktop. **(5 Points)**



5. Create the following AD schema directly off your domain (not part of any previous organizational units) naming usernames the same as their object names and having all passwords be **Htcfun2000**: **(10 Points)**

OU=Final

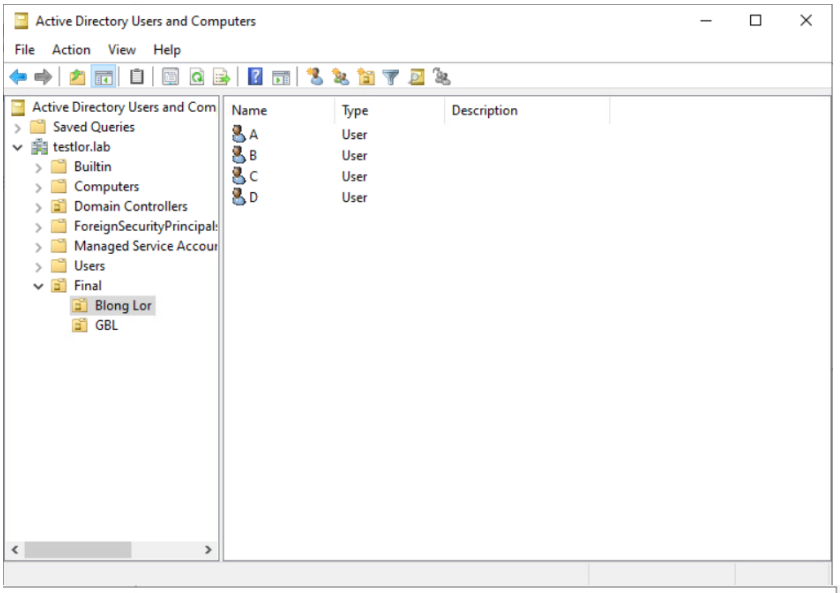
→ OU=***YourFirstandLastName***
- o= A
- o= B
- o= C
- o= D

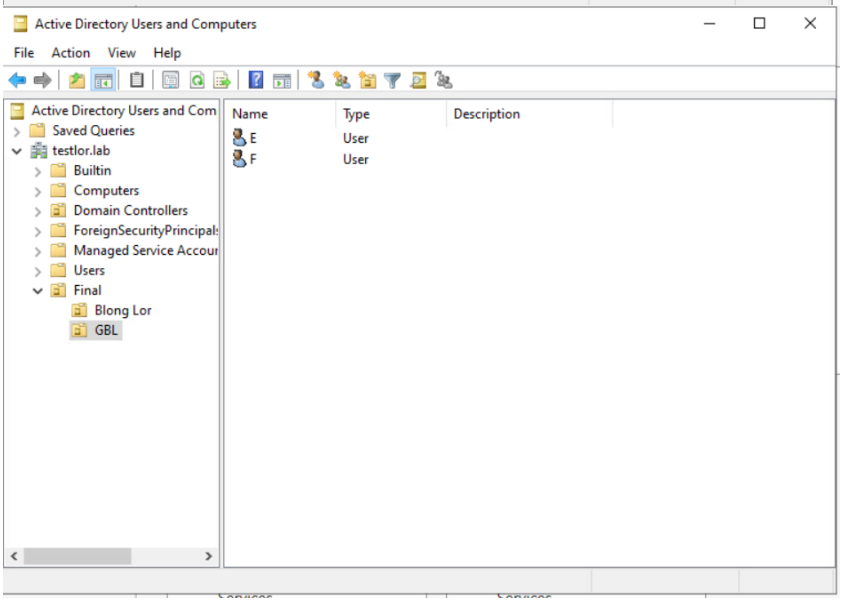
→ OU=***YourComputerName***
- o= E
- o= F

Directions

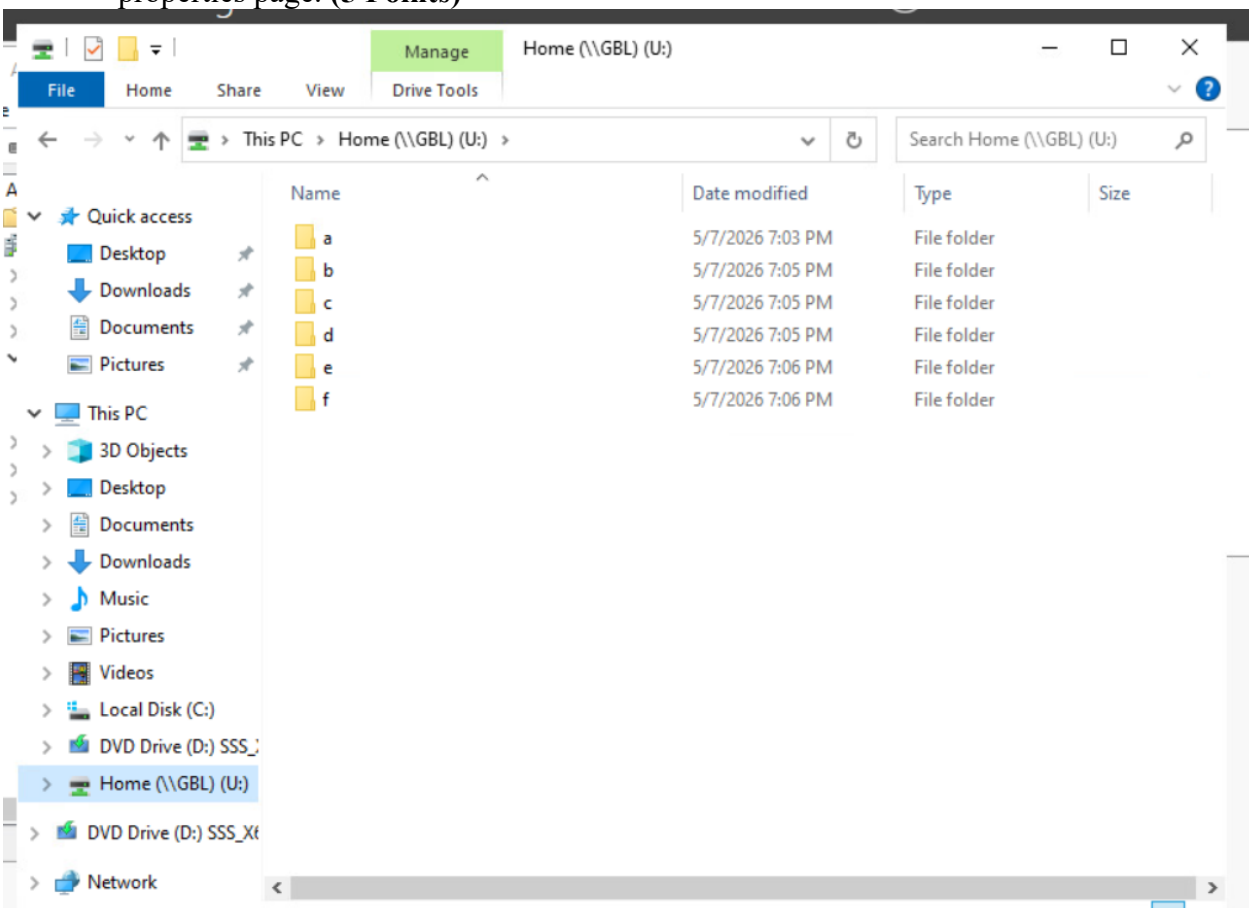
1. Create first the **Final** OU
2. Create the **Yourfirstname** OU inside of the **Final** OU. Then create the users A - D in the **Yourfirstname** OU.
3. Create Y**ourComputerName** OU inside of the **Final** OU. Then create the users E & F in the Y**ourComputerName** OU.

Note: O =User Objects. You may have extra user objects but those listed above must be present

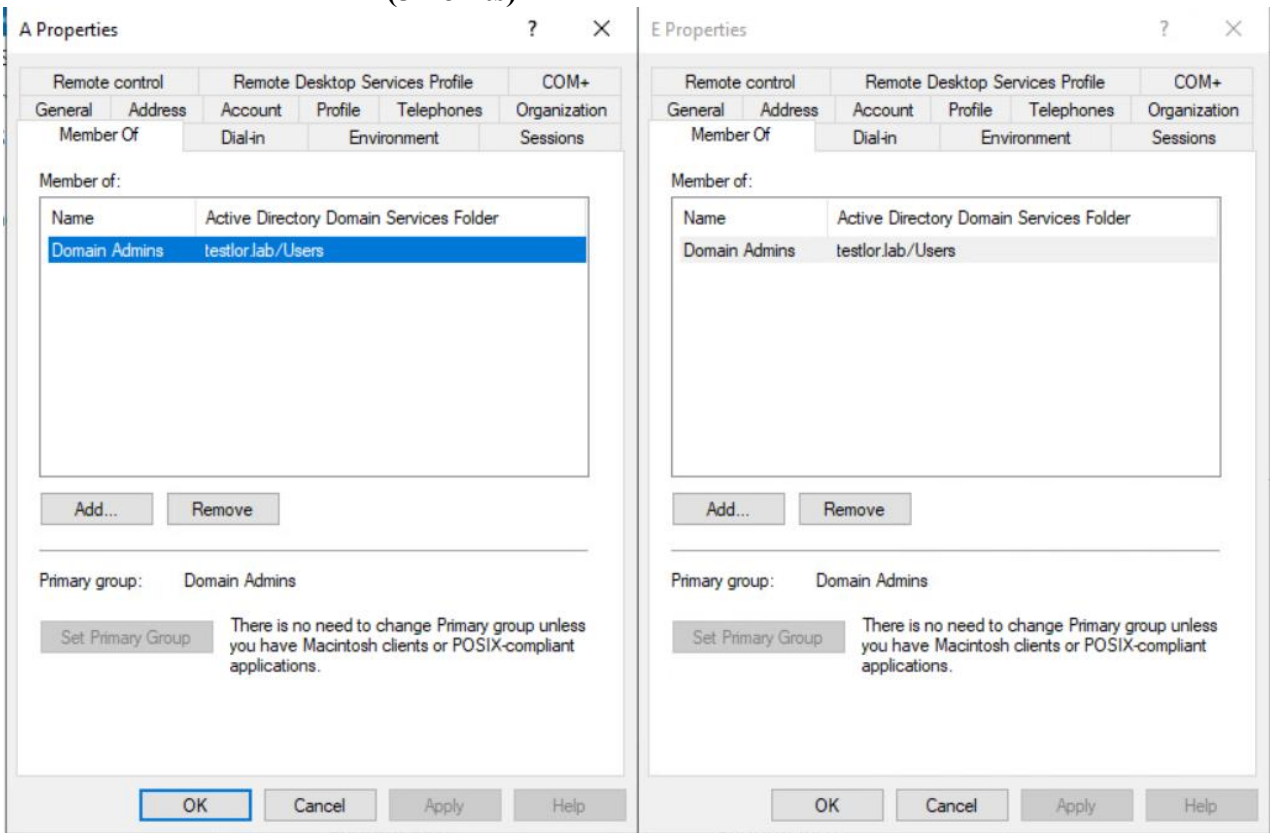




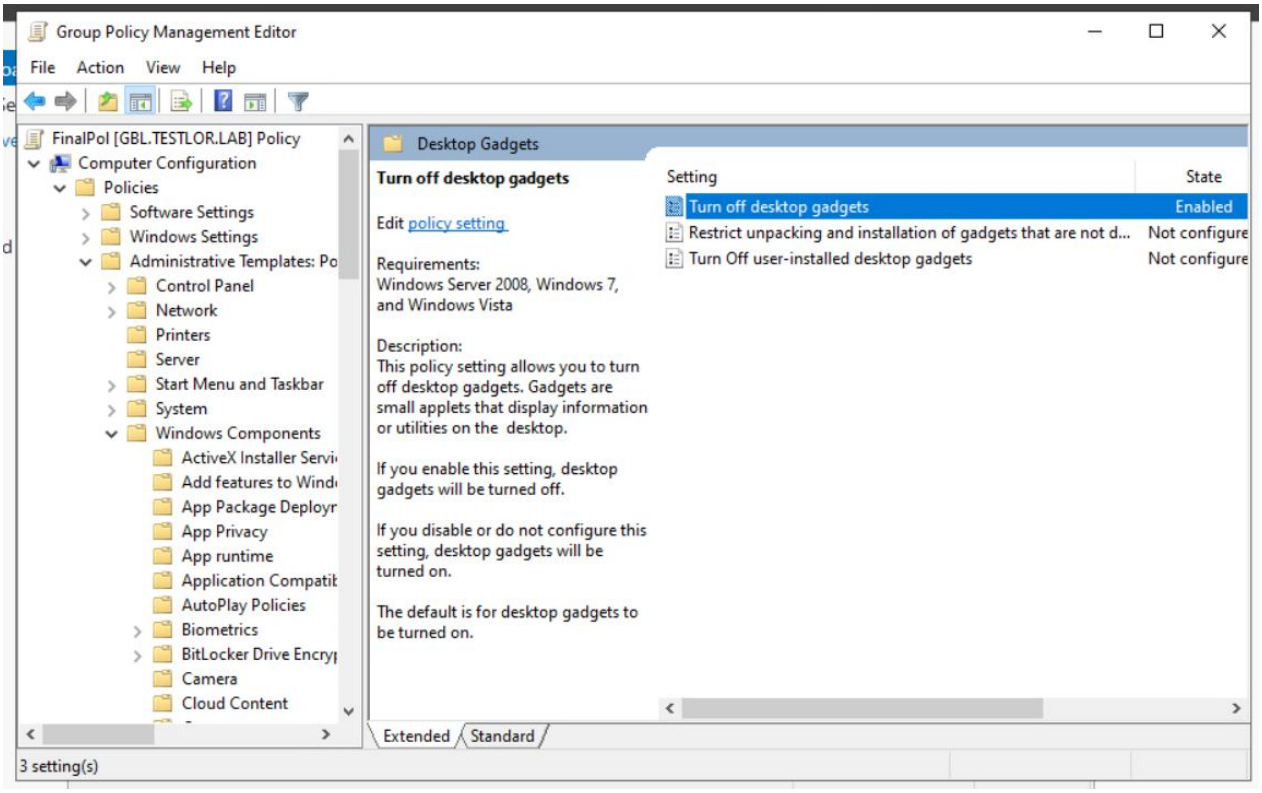
6. Set up the user objects above so that they each have home folders using **Home** as share that you will create off the root of your **C:** partition (remember that you need to create these shares and set up their permissions correctly to work). The home folder option is located in the profile section of the user properties page. **(5 Points)**



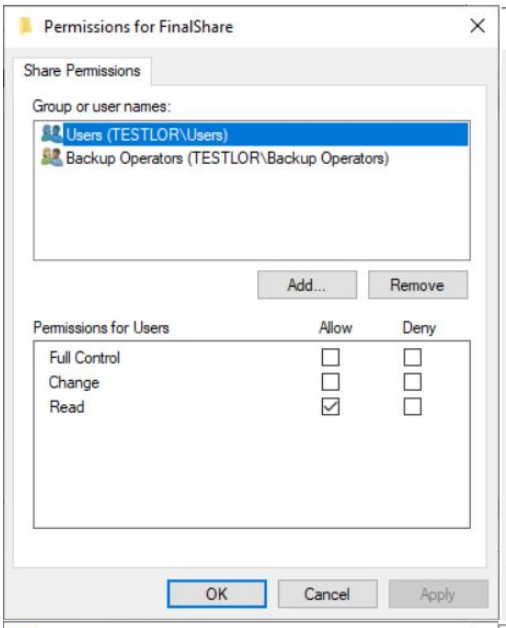
7. Set up the user accounts so that **A** and **E** have **DomainAdmins** group membership and all the others have Domain Users. **(5 Points)**

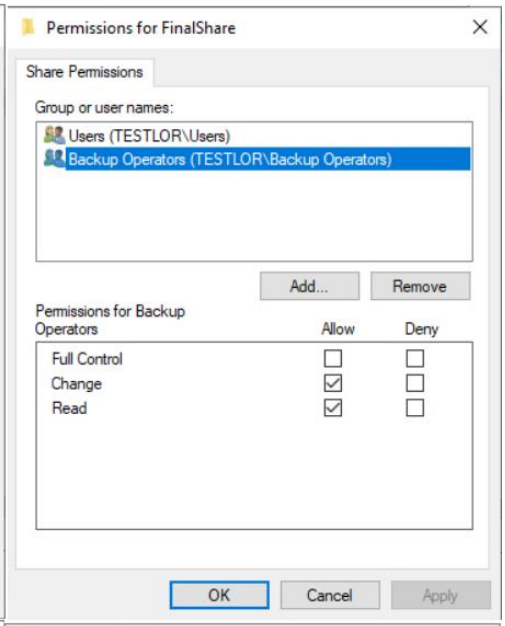


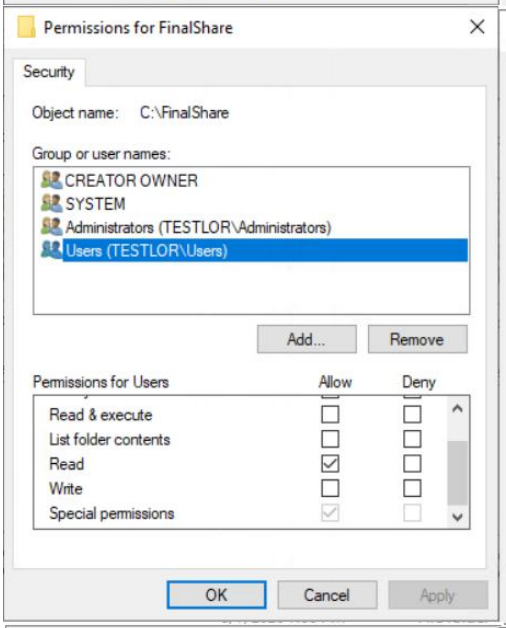
8. Create a new Group Policy called "**FinalPol**" and link it with the Final OU. Then edit it and make sure that the **desktop gadgets** is turned off or disabled *(Hint: Use the Final OU)*. **(5 Points)**

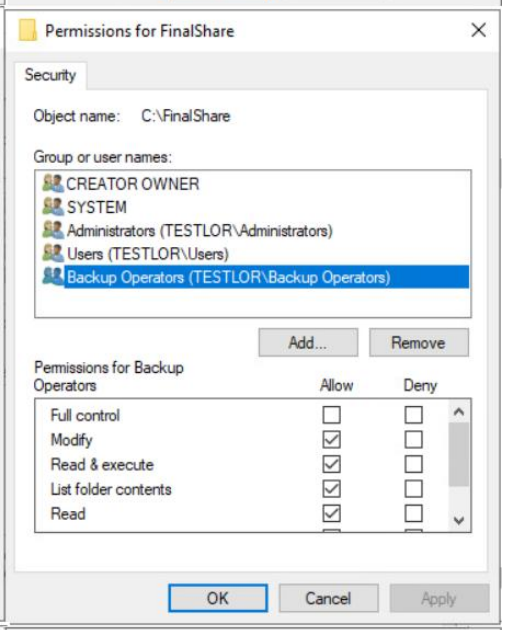


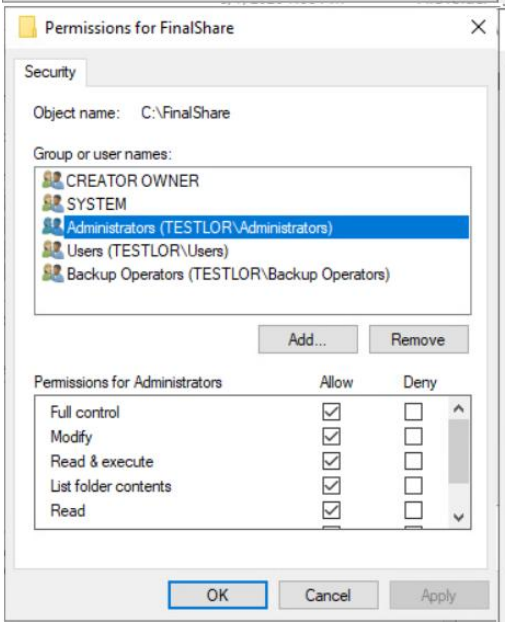
9. Create a share off the root of the **C:** partition called **FinalShare** that will have the following permission requirements; 1. **Share permission** /network access - the **Users** group should have **Read** and **Backup Operators** group should have **Change** permissions. 2. **Security/ NTFS** permissions - **Users** group members should have **Read** (no execute) permission and **Backup Operators** should have **Modify.** 3. **Administrators** Group members should have no restrictions on their access whether local or remote. 4. **No other groups** should have permission to access at all either remote or local (Submit here two screenshots). **(5 Points)**

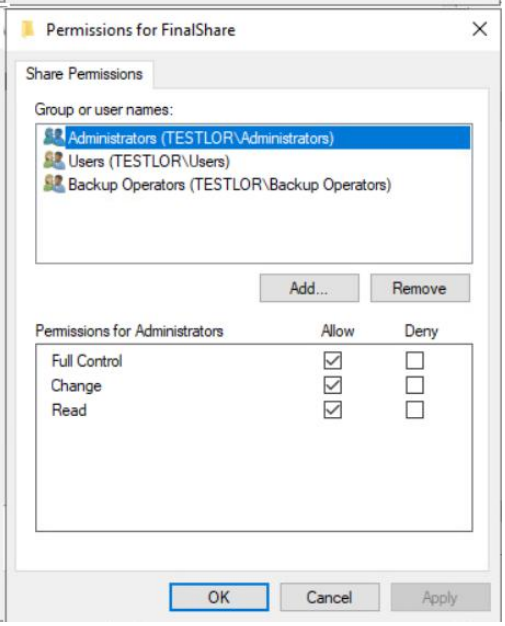

Permissions for FinalShare
Share Permissions
Group or user names:
Users (TESTLOR\Users)
Backup Operators (TESTLOR\Backup Operators)
Add...
Remove
Permissions for Users
Allow
Deny
Full Control
Change
Read
OK
Cancel
Apply


Permissions for FinalShare
Share Permissions
Group or user names:
Users (TESTLOR\Users)
Backup Operators (TESTLOR\Backup Operators)
Add...
Remove
Permissions for Backup Operators
Allow
Deny
Full Control
Change
Read
OK
Cancel
Apply

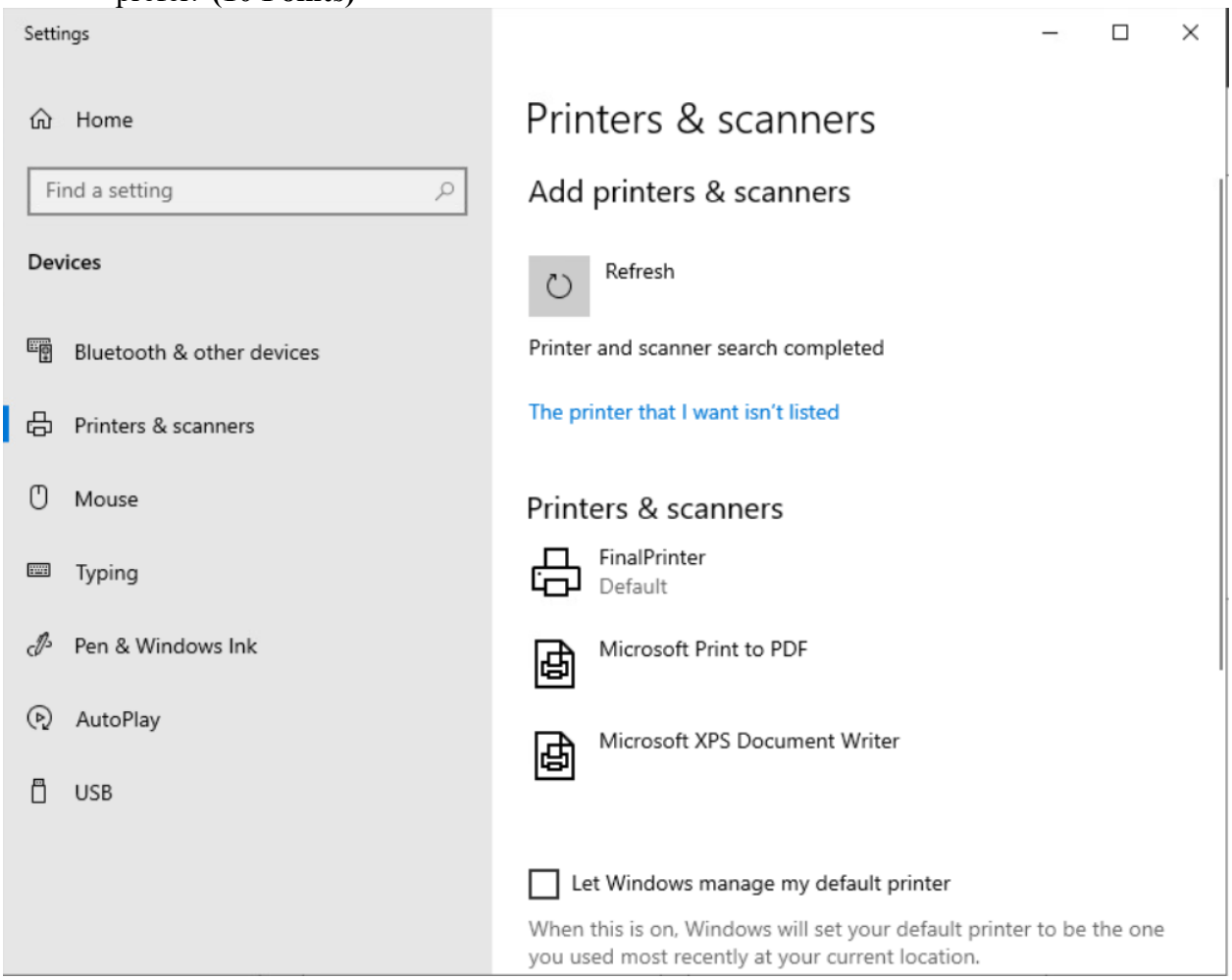

Permissions for FinalShare
Security
Object name: C:\FinalShare
Group or user names:
CREATOR OWNER
SYSTEM
Administrators (TESTLOR\Administrators)
Users (TESTLOR\Users)
Add...
Remove
Permissions for Users
Allow
Deny
Read & execute
List folder contents
Read
Write
Special permissions
OK
Cancel
Apply


Permissions for FinalShare
Security
Object name: C:\FinalShare
Group or user names:
CREATOR OWNER
SYSTEM
Administrators (TESTLOR\Administrators)
Users (TESTLOR\Users)
Backup Operators (TESTLOR\Backup Operators)
Add...
Remove
Permissions for Backup Operators
Allow
Deny
Full control
Modify
Read & execute
List folder contents
Read
OK
Cancel
Apply

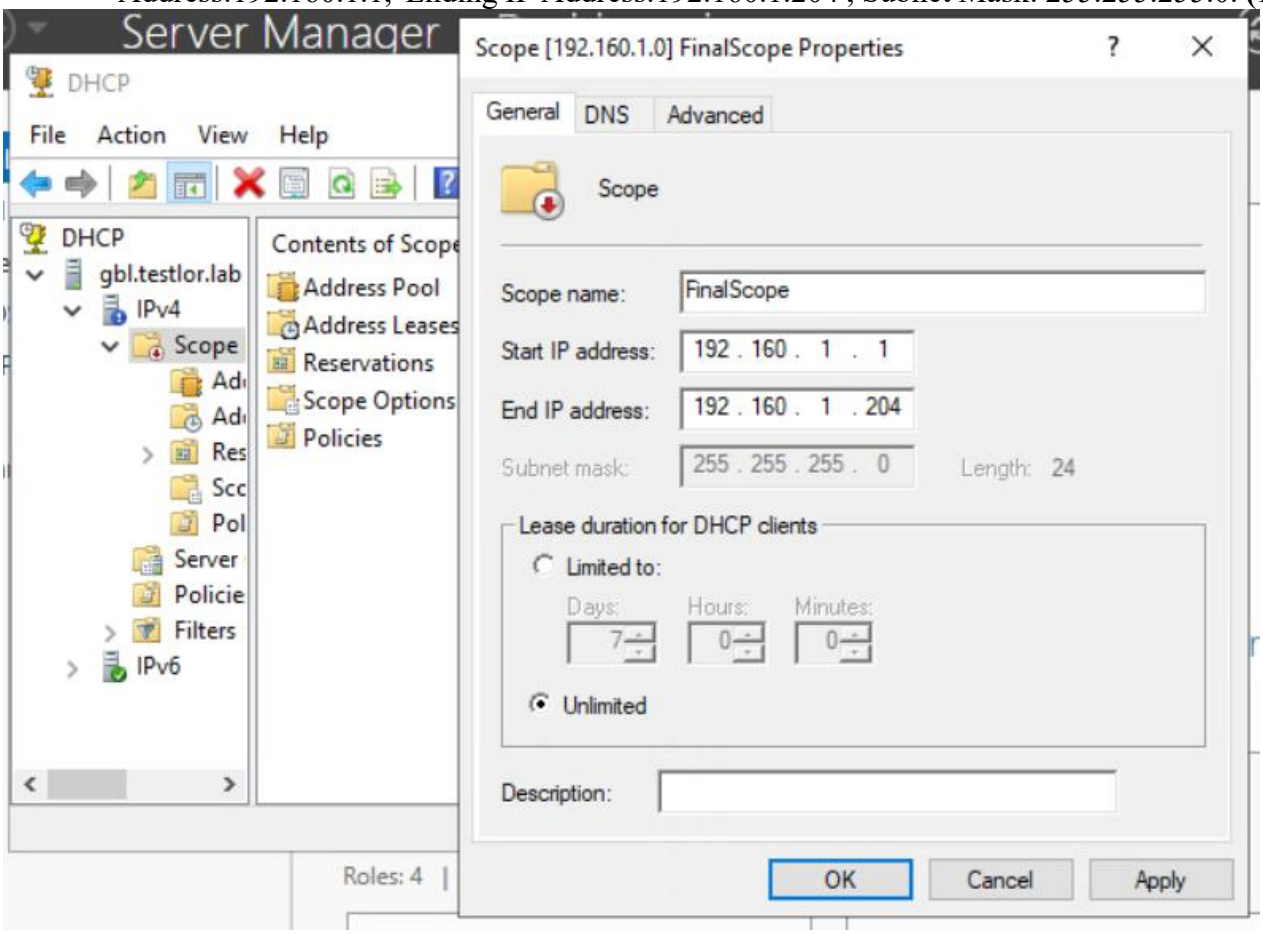

Permissions for FinalShare
Security
Object name: C:\FinalShare
Group or user names:
CREATOR OWNER
SYSTEM
Administrators (TESTLOR\Administrators)
Users (TESTLOR\Users)
Backup Operators (TESTLOR\Backup Operators)
Add...
Remove
Permissions for Administrators
Allow
Deny
Full control
Modify
Read & execute
List folder contents
Read
OK
Cancel
Apply


Permissions for FinalShare
Share Permissions
Group or user names:
Administrators (TESTLOR\Administrators)
Users (TESTLOR\Users)
Backup Operators (TESTLOR\Backup Operators)
Add...
Remove
Permissions for Administrators
Allow
Deny
Full Control
Change
Read
OK
Cancel
Apply

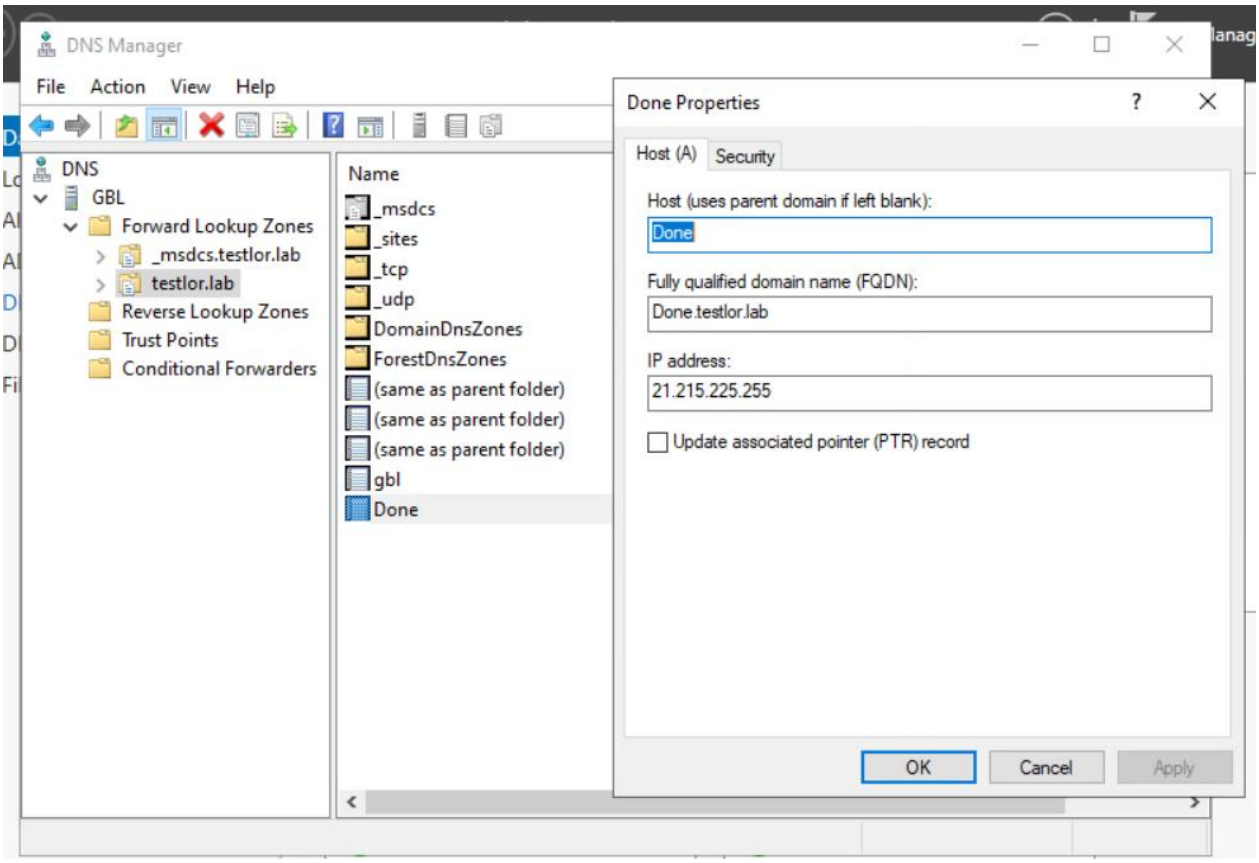
10. Install a fake printer (by any method) and call it **FinalPrinter** and choose any printer model that you prefer. **(10 Points)**



11. Install a DHCP Role. Then add a Scope called ***FinalScope*** with the following info: starting IP Address:192.160.1.1, Ending IP Address:192.160.1.204 , Subnet Mask: 255.255.255.0. **(10 Points)**



12. Create a Host DNS record in your domains zone for a computer called **Done** with an IP address of **21.215.225.255**. **(10 Points)**



Finally, answer the following on this test sheet by circling the BEST answer to each question ***(2 points each)***

13. DNS servers are required to have static IP addresses in a production environment.
    - **a) True**
    - *b)* False

14. Which quota type prohibits users from consuming any disk space beyond what has been defined?
    - a. **hard quota**
    - b. soft quota
    - c. manual quota
    - d. NTFS quota

15. The administrator and Guest user accounts are examples of
    - a. Special identity user account
    - b. Administrative user account
    - c. **Built-in user account**
    - d. Domain Admins group

16. How many drives minimum does it require to have a RAID 5 set?
    - a. One
    - b. Two
    - c. Four
    - d. **Three**

17. Which command-line command can be used to create various object types within Active Directory?
    - a. dsget
    - b. dsquery
    - c. **dsadd**
    - d. dsmove

Total points _____ /100